

PROJECT DESIGN PHASE-II : SOLUTION REQUIREMENTS

SZ METRO TICKET GENERATING SYSTEM IN SERVICENOW

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1. FUNCTIONAL REQUIREMENTS

The functional requirements outline the core capabilities, user interactions, automated scripts, and ticket processing workflows implemented within the ServiceNow platform.

REQ ID	FUNCTIONAL REQUIREMENT (EPIC)	SUB REQUIREMENT / DESCRIPTION (STORY / SUB-TASK)
FR-1	User Registration & Authentication	- Registration through Form (ServiceNow Service Catalog Input) - Registration through Gmail (SSO / OAuth Integration) - Registration through LinkedIn / Enterprise ID
FR-2	User Confirmation & Verification	- Confirmation via Email (Automated ServiceNow Ticket Confirmation) - Confirmation via OTP / SMS Gateway
FR-3	Ticket Booking & Fare Computation	- Station Selection (Dynamic Source & Destination Lookup) - Dynamic Fare Band Calculation (Script Include Matrix Engine) - Passenger Category Selection (Regular, Student, Senior Citizen)
FR-4	QR Digital Ticket Generation & Management	- Dynamic Scannable QR Code Generation on UI Page - Ticket Data Persistence in ServiceNow Custom Table (x_sz_metro_ticket) - Station Manager Scanning & Validation Interface

2. NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements define the system quality attributes, including performance benchmarks, security protocols, reliability, usability, and availability standards.

REQ ID	NON-FUNCTIONAL REQUIREMENT	DESCRIPTION
NFR-1	Usability	Intuitive and responsive ServiceNow Service Portal interface allowing passengers to complete ticket generation in under 3 steps.
NFR-2	Security	Encrypted QR code payloads, Role-Based Access Control (RBAC) via ServiceNow ACLs, and secure OAuth2 authentication for external logins.
NFR-3	Reliability	Guaranteed data consistency across ticket table records with automated transaction logging and error handling in Script Includes.
NFR-4	Performance	Instant fare computation and dynamic QR rendering with response times under 1.5 seconds during peak booking transactions.
NFR-5	Availability	High availability guaranteed by ServiceNow Cloud Infrastructure with 99.9% uptime for continuous ticketing access.
NFR-6	Scalability	Capable of handling concurrent passenger ticket generation requests during rush hours without degradation of portal response times.